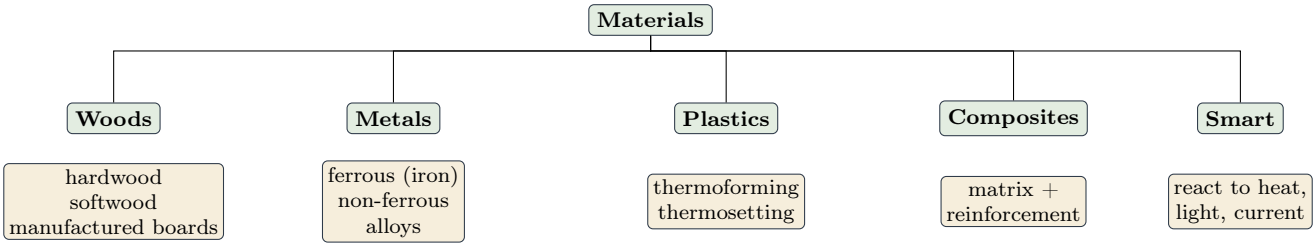
Materials: the minimum you need

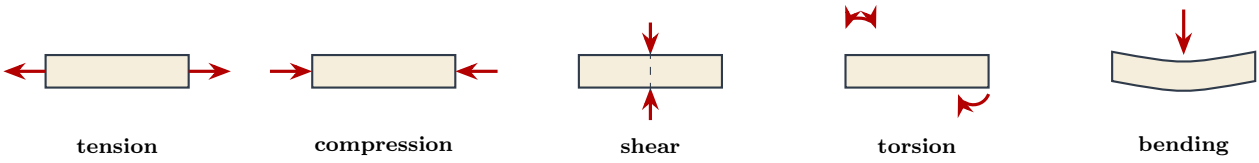
IGCSE D&T 0445 · Grade 11 · Paper 4 Section A and the M2 quiz

Start with properties, then use them to compare. Every "Exam" note comes from a real mark scheme; the paper code (for example M/J 24 P42 Q3) tells you where to find it.

1. Start here: properties



Property	What it means	Where you meet it
Tensile strength	resists being pulled apart	cables, ties, steel bars in concrete
Compressive strength	resists being squeezed	concrete, pillars, brick
Shear strength	resists layers sliding past each other	bolts, rivets, pins
Torsional strength	resists twisting	shafts, pylons
Bending strength	resists bending under a load	beams, shelves
Hardness	resists scratching, dents and wear	cutting tools, worktops
Toughness	absorbs a sudden blow without breaking (opposite: *brittle*)	hammers, helmets
Ductile	can be stretched or drawn into a wire	copper wire, steel cable
Malleable	can be hammered or pressed into shape	sheet metal, car panels
Elasticity	springs back to its shape after bending	springs, flexible plywood
Durable	lasts in use, resists weather and wear	window frames
Corrosion resistance	does not rust or react with its surroundings	stainless steel, copper roofs
Conductor / insulator	lets electricity or heat pass / blocks it	copper wire / PVC, rubber



How marks are lost (checked against the mark schemes)

- **"Strong" alone scores nothing.** Say *strong in tension* or *strong in compression* (M/J 22 P42 Q1a, M/J 25 P42 Q2).
- **Cost is not accepted** unless you justify it, and "do not allow cost related benefits" appears often (M/J 22 P41 Q2b, M/J 24 P41 Q2a, O/N 23 P41 Q1c).
- **Name the material when asked.** "Plastic" is not an insulator; PVC, rubber or ceramic is (M/J 24 P41 Q7). A conductor can be any named metal (M/J 22 P41 Q8).
- **Some properties are refused for steel.** For cables and pylons "corrosion resistant" and "malleable" were not allowed, because plain steel rusts (M/J 25 P42 Q2, O/N 22 P42 Q3).
- **One mark, one point.** A 3-mark question wants three different properties, not one property said three ways.

Compare two materials in three moves

1. Name the property the job needs (window frame: weather resistance, low maintenance). **2.** Say which material has it and how it compares (plastic does not need repainting, wood does). **3.** Give the consequence for the product (the frame lasts without upkeep). Two marks usually means two such points.

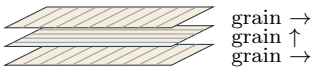
2. Woods

Hardwood and softwood describe the tree, not how hard the wood is. Softwood comes from fast-growing coniferous trees (cones, needles); hardwood from slower-growing deciduous trees. So hardwood is usually dearer. Always season timber (dry it, often in a kiln) or it shrinks and warps after you have built with it.

Material	Type	Key properties	Typical use
Pine, spruce	softwood	easy to work, low cost, knotty	construction, simple furniture, joinery
Oak	hardwood	strong, tough, durable; attacks steel screws	quality furniture, interior woodwork
Beech	hardwood	hard, close-grained, strong; prone to warping	furniture, toys, tool handles
Plywood	manufactured board	layers glued with grain at 90°: strong, hard to split	furniture, boat hulls (marine grade)
MDF	manufactured board	smooth, easy to machine, takes paint; no grain	indoor furniture, flat-pack
Chipboard	manufactured board	chips and resin; cheaper, weaker	indoor furniture, usually veneered
Hardboard	manufactured board	thin, smooth one side; weak	cupboard backs (non-structural)

Boards against natural timber: available in big sheets, uniform with no knots, do not split, use waste wood. Natural timber is better when you need long, strong members, or a quality appearance.

Defects the mark scheme asks for: knots, shakes (splits), twisting, warping (M/J 22 P42 Q9, M/J 24 P41 Q10).



plywood: crossed layers

3. Metals

Ferrous metals contain iron: they are magnetic, have high melting points and rust (stainless steel is the exception). **Non-ferrous** metals contain no iron, so they do not rust and are not magnetic. An **alloy** is a mixture of metals made to improve properties (brass, steel, duralumin). *An alloy is not a composite.*

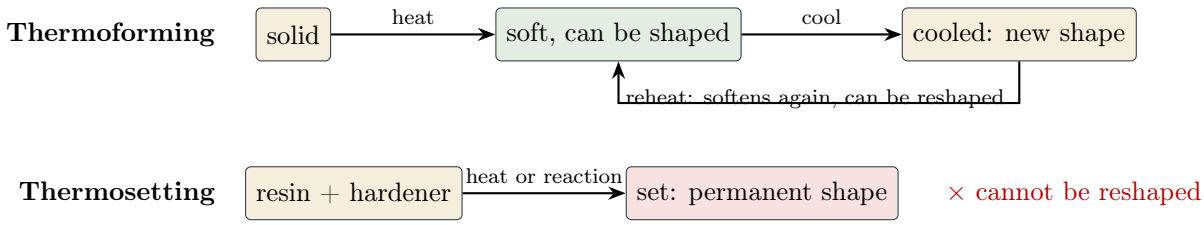
Metal	Group	Key properties	Typical use
Mild steel	ferrous alloy	tough, ductile, malleable, strong in tension; rusts	construction, nuts and bolts, car panels
High carbon steel	ferrous alloy	hard and tough but more brittle; can be heat-treated	chisels, saws, hand tools
Stainless steel	ferrous alloy	hard, corrosion and wear resistant, hygienic	cutlery, kitchen and surgical equipment
Cast iron	ferrous	brittle, rusts; used for casting	manhole covers, brake discs
Aluminium	non-ferrous	light (low density), does not rust	ladders, window frames, bicycles
Duralumin	non-ferrous alloy	very strong, hard, lightweight	aircraft, boat building
Copper	non-ferrous	very ductile, excellent conductor; goes green (verdigris)	wire, plumbing, roofs
Brass	non-ferrous alloy	golden, machines well; copper + zinc	plug pins, handles, instruments

Say it like this (steel in a structure, M/J 22 P42 Q1a, M/J 25 P42 Q2)

Accepted points for mild steel: *strong in tension*, *resists compression*, *resists torsion / bending*, *ductile*, *tough*, *easily joined* (welding, bolts), *can be recycled*. Not accepted on their own: strong, rigid, flexible, cheap.

4. Plastics

Plastic means polymer: long chain molecules, mostly made from oil. The one distinction that matters is what heat does to it.



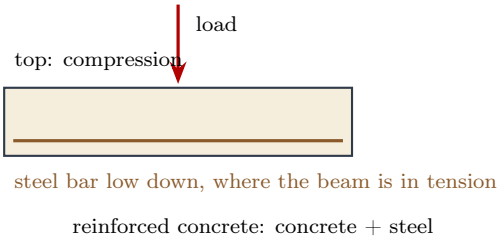
Plastic	Type	Key properties	Typical use
Acrylic (PMMA)	thermoforming	hard, clear (more transparent than glass), UV and weather resistant	signs, light lenses, aircraft canopies
PVC	thermoforming	good tensile strength, chemical and water resistant, reasonably impact resistant	pipes, cable insulation, window frames
Nylon	thermoforming	low friction, resists abrasion, self-lubricating	gears, bearings, bushes, rope
HDPE / LDPE	thermoforming	chemical resistant; LDPE is the cheaper grade	bottles, bins, packaging film
Polypropylene (PP)	thermoforming	flexible, resists stress, chemical and water resistant	food boxes, ropes, chemical tanks
ABS	thermoforming	durable, weather and chemical resistant, impact resistant	appliance housings, casings for power tools
Epoxy resin	thermosetting	high adhesive strength, electrical insulator	adhesives, circuit boards, coatings
Polyester resin (GRP)	thermosetting	stable, low cost, easy to use with glass fibre	boat hulls, car body panels
Urea, melamine, phenol formaldehyde	thermosetting	hard surface, heat and chemical resistant, electrical insulators	plugs, switches, worktops

Plastic against wood, a favourite (M/J 22 P41 Q3, O/N 22 P42 Q1)

Plastic is better: weather resistant, low maintenance, can be moulded or extruded to shape, lighter, stable in size. **Worse:** can discolour in sunlight, can become brittle, less compression resistance, oil-based and harder to recycle. **Glass over plastic bottles (environment):** recyclable and reusable, not made from oil. **Plastic over glass (function):** lighter, does not break, easily moulded.

5. Composites and smart materials

A **composite** combines two or more materials that keep their own identity: the **matrix** surrounds and binds, the **reinforcement** (fibre) carries the load. You can usually see the parts. That is why bronze (an alloy) is not a composite.



Composite	Why it is used
Reinforced concrete	concrete resists compression, steel resists tension
GRP	glass fibre in polyester resin: light, strong, moulds to any shape (boat hulls)
CFRP	carbon fibre in resin: best strength to weight, expensive (bikes, F1, aircraft)
Kevlar	aramid fibre, stronger than steel, light and flexible when woven (protective clothing)
Plywood	laminated timber layers: strong, resists splitting

Smart and modern materials react to their surroundings: **thermochromic** inks change colour with temperature (forehead thermometer strips); **shape memory alloy** (nitinol) returns to its shape when heated or when current passes through it (stents, spectacle frames); **shape memory polymer** deforms with voltage or temperature; **Polymorph** pellets fuse at 62°C and can be moulded by hand, then set tough (prototyping handles and parts).

Structural materials in one line each (M/J 24 P41 Q2a, O/N 23 P41 Q1c, O/N 25 P41 Q1)

Wood: renewable, easy to work, good appearance. **Steel:** resists most forces, formed into many sections. **Concrete:** strong in compression, can be cast on site, durable; *weak in tension* unless reinforced; slow to cure. **Composite:** properties are designed for the job and avoid long timber lengths. Named composites accepted: reinforced concrete, plywood, GRP, carbon fibre, Kevlar.

6. From the need to the material

The job needs...	Think of...
resist being pulled (cable, tie)	steel (ductile, strong in tension); Kevlar, carbon fibre
resist being squeezed (pillar, foundation)	concrete, steel; hardwood
light and strong	aluminium, duralumin, GRP, CFRP
no rusting outdoors or in food areas	stainless steel, aluminium, copper, most plastics
moulded into a complex shape	thermoforming plastics (injection moulding), GRP, cast metals
electrical conductor / insulator	named metal, copper / PVC, rubber, ceramic, nylon
low friction moving part	nylon (self-lubricating), bronze bearings
a renewable or recyclable choice	wood is renewable; steel, aluminium, glass recycle well; oil-based plastics are the concern

Sustainable resource (M/J 24 P42 Q10b): one that can renew itself, such as managed forest timber. Do not confuse it with *recyclable*.

7. Try the real questions

Paper	Question	Marks
M/J 24 P41, P43 Q2a	Give one benefit of each: wood, steel, concrete, composite	4
M/J 24 P42 Q3	Link six materials (hardwood, aluminium, PVC, GRP, concrete, stainless steel) to their properties	4
M/J 22 P41, P43 Q3	Two advantages and two disadvantages of plastic against wood for window frames	2+2
M/J 25 P42 Q2	Two properties of steel that suit suspension bridge cables	2
O/N 22 P42, P43 Q3	Three reasons steel suits electricity pylons	3
O/N 25 P41–43 Q1	Name a composite for an edging block and another composite used in a structure	1+1
M/J 24 P41, P43 Q7	Name one specific electrical insulating material	1
O/N 22 P42, P43 Q1	Environmental reason for glass bottles; functional reason for plastic bottles	1+1

Papers from the same session share most Section A questions, so one question appears in two or three variants. Past papers are in `class_materials/past_papers_0445`; mark schemes are the authority for what scores.